

- 7 (a) Neutron **A1**
 $x = 4$ **A1**
- (b) (i) The binding energy of a nucleus is the minimum energy required to completely separate the nucleus into its constituent nucleons (protons and neutrons) to infinity. **B1**

MC:	<i>Generally very poorly done. Most students were lacking in keywords in their definitions</i>
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(ii) $E_{\text{released}} = BE_{\text{products}} - BE_{\text{reactants}}$
 $= 139 (8.39) + (5 (8.74) - 235 (7.60))$ **C1**
 $= 210.51 \text{ MeV}$
 $= 3.36816 \times 10^{-11} \text{ J}$
 $= 3.37 \times 10^{-11} \text{ J}$ **A1**

MC:	<i>Generally not very well done. Most students missed out that the information given is the binding energy <u>per nucleon</u>, and hence did not multiply the mass numbers in their workings.</i> <i>A small handful had erroneous conversion of MeV into J.</i>
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(iii) $(\text{loss in mass}) c^2 = E_{\text{released}}$
 $(\text{loss in mass}) (3.00 \times 10^8)^2 = 3.36816 \times 10^{-11}$ **C1**
 $(\text{loss in mass}) = 3.74 \times 10^{-28} \text{ kg}$ **A1**

MC:	<i>Generally well done. Ecf was given should their answer in (ii) be incorrect.</i>
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- (iv) When more energy is being released in a nuclear reaction, more stable daughter nuclei are formed as a result. **B1**
 Since reaction 1 releases more energy than reaction 2, **B1**
 reaction 1 is more likely to occur. **B1**

MC:	<i>Generally well done. Ecf was given should their answer in (ii) be incorrect. Students' conclusion and explanations should be adjusted according to their answer in (b)(ii). (i.e. if the energy released obtained in (ii) is lower than reaction 2, students must conclude that reaction 2 is more likely to occur to attain credit)</i> <i>A small handful of students mixed up the concepts of spontaneity of reactions with the likelihood of reactions. Students are advised not to haphazardly introduce scientific terms outside of what the question has stated or is required to answer the question to avoid being marked down unnecessarily.</i>
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